Curate for Attention, Sample for Convolutions: Taming False Alarms in Continuous Edge AIoT Sensing

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***Abstract*—Deep learning detectors deployed in continuous edge Artificial Intelligence of Things (AIoT) sensing face extreme temporal background dominance: over 80% of operational frames contain no target objects. Training with negative (background-only) frames suppresses false alarms, yet prior work evaluates negative-to-positive ratios for single architectures in isolation, leaving open whether an optimal ratio transfers across designs. We present a cross-architecture empirical study of negative-frame ratio sensitivity across three matched-capacity (≈ 2.6M parameters) lightweight edge detectors—YOLO11n, YOLO26n, and YOLO12n—spanning convolutional, hybrid, and attention-centric feature aggregation. Sweeping five nested ratio configurations (0%–80% negatives) on a 15,723-frame in-cabin driver-monitoring corpus, we find that the direction of false-alarm suppression is consistent across all three architectures, but the magnitude and the optimal accuracy–suppression trade-off are architecture-dependent. At 80% negatives, all three detectors converge to a narrow false-positive band (4–9 raw detections per 1,272 negative test frames) despite up to 31.5$\times$ variation in their respective suppression trajectories—indicating that negative-frame prevalence, not architecture or inference latency, is the dominant lever for false-alarm control in edge AIoT deployments.**

***Index Terms*—Artificial Intelligence of Things (AIoT), edge intelligence, driver monitoring systems, object detection, negative frames, hard-negative mining, class imbalance, false-positive rate, YOLO, area attention.**

I. Introduction

Object detectors deployed in edge AIoT ecosystems (including in-cabin driver monitoring systems, robotic platforms, and persistent visual surveillance nodes) operate under extreme temporal class imbalance [1]. In vehicular sensing, a camera stream processes video continuously across multi-hour trips, yet safety-critical driver-cue events (e.g., mobile phone usage, drinking, yawning, hand-over-mouth distraction) occupy only a minor fraction of the visual stream [12]. Under natural operational distributions, over 80% of encountered frames contain no target objects.

This persistent background dominance implies that an edge detector's operational utility is governed not merely by mAP over object-dense benchmarks, but primarily by its ability to maintain discriminative silence on background frames. Detectors exhibiting non-trivial false-positive rates induce nuisance alerts that precipitate driver alert fatigue, waste edge compute cycles, trigger unwarranted cellular uplinks in distributed IoT fleets, and exhaust thermal envelopes on resource-constrained systems-on-chip (SoCs) [12], [13].

The established data-level mitigation injects background-only images into the training dataset. Practitioner heuristics recommend 0–10% background imagery [8], while domain studies indicate that omitting negative frames degrades open-world false-alarm resilience [5], [6]. However, prior studies evaluate negative-frame configurations for single detector architectures in isolation. It remains unknown whether the optimal negative ratio (or the utility of hard-negative curation) is an intrinsic domain property or depends on the architectural feature-aggregation design. If negative-frame sensitivity is architecture-dependent, edge model selection and dataset curation cannot be decoupled.

We present a controlled cross-architecture investigation on a real-world driver-monitoring dataset, evaluating three lightweight detectors with matched model capacity ($\approx$ 2.6M parameters): YOLO11n [9], YOLO26n [10], and YOLO12n [11]. These models span three distinct spatial aggregation designs: hybrid convolutional-attention (YOLO11n), pure reparameterized feedforward convolution (YOLO26n), and area-attention (YOLO12n). Using a 15,723-frame corpus with 80.9% natural negative prevalence, we address two research questions:

- **RQ1 (Ratio Sensitivity):** How does negative-frame scaling affect false-positive suppression across architectures, and does the ratio optimizing detection accuracy differ between convolutional and attention-based designs?
- **RQ2 (Curation Quality):** At a compute-efficient reference ratio ($r_{\mathrm{ref}} = 40\%$), does substituting random negatives with hard-mined negatives yield further gains at matched dataset cardinality?

Our contributions: (1) the first cross-architecture negative-ratio sensitivity sweep across three matched-capacity edge detectors on a safety-critical AIoT dataset, examining whether curation configurations transfer across paradigms; (2) a controlled hard-negative mining protocol benchmarked against random sampling at matched cardinality (r_{ref}=40%), isolating sample entropy from volume; (3) an empirical latency benchmark (16.8–21.7 ms) coupled with an analytical mapping from $\mathrm{FP}/1\mathrm{k}$ to hourly nuisance alerts ($\mathcal{A}_h$), illustrating that data curation governs edge reliability independently of inference latency.

II. Related Work

Class Imbalance and Negative Sampling. Foreground-background imbalance is canonical in detection [1]. Focal Loss [1] downweights easy negatives via $(1 - p_t)^\gamma$; OHEM [14] constructs batches from high-loss regions; ATSS [3] showed that assignment strategy matters more than anchor design. Wang *et al.* [2] demonstrated negative representation drift under iterative backpropagation, and SNIPER [4] confirmed that false-positive background chips improve precision without multi-scale overhead.

Lightweight Edge Detectors. Edge AIoT mandates compact architectures under strict latency and memory constraints [15]. YOLO models [9], [10] employ decoupled heads, Task-Aligned Assigners, and Distribution Focal Loss (DFL) [16]. YOLO11 couples convolutional feature pyramids with neck-level partial self-attention (C2PSA). YOLO26 introduces structural reparameterization blocks (RepConv) to optimize feedforward edge inference without attention branches. YOLO12 [11] introduces an attention-centric design via Area Attention (A2C2f), computing self-attention over partitioned horizontal and vertical visual areas with linear complexity. Benchmarking these three 2.6M-parameter variants isolates feature aggregation mechanisms while keeping parameter volume strictly controlled.

Negative-Frame Training in AIoT. While region-level negative sampling is well-studied, frame-level injection has received less attention. Beery *et al.* [5] showed models trained without empty backgrounds suffer generalization penalties; Felzenszwalb *et al.* [6] showed that bootstrapping with background scenes suppresses false detections; Gupta *et al.* [7] found scaling negative pairs to 1000:1 essential for human-object interaction. In driver monitoring systems (DMS), datasets [18], [19] emphasize positive diversity, but deployed nodes encounter vast negative stretches [12], [17]. Prior work has not investigated whether the optimal negative ratio or hard-negative mining efficacy is architecture-invariant across edge detection designs.

III. Methodology

A. Problem Formulation

Let $\mathcal{S} = \{X_t\}_{t=1}^{T}$ denote a continuous streaming video feed captured by an edge camera sensor, where $X_t \in \mathbb{R}^{H\times W\times 3}$ represents the image frame at time step t. Each frame X_t is associated with ground-truth $Y_t = \{(c_k, b_k)\}_{k=1}^{K_t}$, with class $c_k \in \{1,\ldots,C\}$ and bounding box $b_k \in [0,1]^4$. We partition frames into disjoint categories:

$$\mathcal{D}_{\text{pos}} = \{(X_t, Y_t) \mid K_t \geq 1\}, \quad \mathcal{D}_{\text{neg}} = \{(X_t, \emptyset) \mid K_t = 0\}. \tag{1}$$

Given a detector parameterized by θ, inference on frame X yields predicted tuples $f_\theta(X) = \{(\hat{c}_j, \hat{b}_j, s_j)\}_{j=1}^{M}$. At confidence threshold $\tau \in (0,1]$, the filtered prediction set is:

$$\hat{\mathcal{Y}}_\tau(X) = \left\{(\hat{c}_j, \hat{b}_j, s_j) \in f_\theta(X) \;\middle|\; s_j \geq \tau\right\}. \tag{2}$$

For any negative frame $X \in \mathcal{D}_{\text{neg}}$, the ground-truth is empty ($Y{=}\emptyset$), so every detection in $\hat{\mathcal{Y}}_\tau(X)$ constitutes an operational false positive. The deployment cost metric on the held-out negative test set $\mathcal{D}_{\text{test}}^{\text{neg}}$ is:

$$\text{FP/1k} = \frac{1000}{|\mathcal{D}_{\text{test}}^{\text{neg}}|} \sum_{X\in\mathcal{D}_{\text{test}}^{\text{neg}}} |\hat{\mathcal{Y}}_\tau(X)|. \tag{3}$$

In an operational edge deployment at frame rate f_{FPS} over one hour ($T_{\text{hr}}{=}3600\,\text{s}$), with operational negative prevalence $p_{\text{neg}} \in (0,1]$, the expected hourly nuisance alert rate is:

$$\mathcal{A}_h = \frac{T_{\text{hr}}}{1000} \times f_{\text{FPS}} \times p_{\text{neg}} \times \text{FP/1k} = 3.6\, f_{\text{FPS}}\, p_{\text{neg}}\, \text{FP/1k}. \tag{4}$$

Under unconditioned driving streams ($p_{\text{neg}} \approx 0.809$), (4) models the real-world nuisance alert burden experienced by the driver and the edge communication stack. In our benchmark, operational false alarms (FP/1k and $\mathcal{A}_h$) are measured across all 1,272 held-out negative test frames at operational confidence threshold $\tau = 0.25$ and non-maximum suppression (NMS) IoU $= 0.70$, matching the hard-negative mining threshold in Section III-E. Detection accuracy metrics (mAP_{50} at IoU $= 0.50$, $\text{mAP}_{50:95}$ averaged over IoU $\in [0.50, 0.95]$, precision P, recall R) are evaluated across the full test set over the confidence sweep, with P and R reported at the maximum-F_1 operating point. Note that P and R (maximum-F_1 threshold) and FP/1k (fixed $\tau{=}0.25$) reflect different operating points; both are reported in Table III but should not be assumed to describe a single deployable configuration.

B. Architectural Loss Dynamics on Negative Frames

All three evaluated models employ dense anchor-free Task-Aligned Assignment. On an unannotated negative frame ($Y{=}\emptyset$), the assigner matches zero positive anchors across all pyramid scales $s \in \{P_3, P_4, P_5\}$. As a result, bounding-box regression loss ($\mathcal{L}_{\text{box}}$) and distribution focal loss ($\mathcal{L}_{\text{dfl}}$) vanish to zero identically. The overall loss collapses strictly to classification binary cross-entropy (BCE) accumulated across all spatial anchor locations:

$$\mathcal{L}_{\text{det}}^{\text{neg}} = -\lambda_{\text{cls}} \sum_{s=1}^{S} \sum_{i=1}^{H_s W_s} \sum_{c=1}^{C} \log\left(1 - \hat{p}_{s,i,c}\right), \tag{5}$$

where $\hat{p}_{s,i,c}$ is the predicted probability for class c at grid cell i of scale s. Without negative training frames ($r{=}0\%$), classification gradients on background regions arise solely from non-target patches within positive frames.

Although the loss formulation is mathematically shared, the backpropagation pathways through the feature extractors differ, and we hypothesize this leads to architecture-dependent responses to negative-frame training:

- **YOLO26n (Pure Reparameterized CNN):** Negative gradients propagate strictly through local convolutional receptive fields. We hypothesize that weight updates must attenuate spatial filters responding to high-frequency edge and texture patterns resembling driver cues (e.g., steering wheel rims, seatbelt fixtures).
- **YOLO11n (Hybrid CNN-Attention):** Negative gradients update both convolutional stages and neck self-attention projection weights (C2PSA). This may enable

- cations compared to purely local filters.

broader spatial modulation across prominent feature locations compared to purely local filters.

- **YOLO12n (Area Attention):** Negative gradients pass through Area Attention modules (A2C2f) that compute self-attention across partitioned visual areas. We hypothesize that this dynamic receptive field may suppress extended background structures (such as headrests or window reflections) more holistically than local convolutions, though we have not verified this via attention visualization or targeted ablation.

This architectural divergence motivates our inquiry into whether optimal negative-frame ratios and hard-negative mining efficacy differ across detection designs.

C. Dataset and Partitioning

Experiments utilize a 15,723-frame in-cabin DMS corpus captured across diverse driver subjects, cabin illumination conditions, vehicle cabins, and camera viewpoints. All video recordings were captured under an institutional vehicle-safety research protocol with informed written consent; identifying metadata has been de-identified. The dataset contains 3,001 positive frames with annotated driver cues across four safety-critical categories: (1) `phone_use` (2,437 frames, 81.2%, manual device interaction or holding to ear), (2) `drinking` (264 frames, 8.8%, beverage container near mouth), (3) `yawning` (159 frames, 5.3%, observable fatigue), and (4) `hand_over_mouth` (141 frames, 4.7%, facial occlusion). All positive frames are annotated with tight 2D bounding boxes enclosing target cues and quality-verified by consensus review (single primary cue per positive frame). The remaining 12,722 frames contain no target cues, yielding a natural negative prevalence of 80.9%. All frames are partitioned using a stratified 80/10/10 split (seed=42): validation and test benchmarks contain 1,572 frames each (300 pos, 1,272 neg; 80.9% negative prevalence), while the training universe comprises 2,401 positive frames (stratified across all four categories) and a candidate pool of 10,178 negatives. Validation and test benchmarks are held strictly fixed across all conditions. Evaluating on a natural-distribution test set preserves the background-dominant reality necessary to quantify $\mathrm{FP}/1\mathrm{k}$; an artificially rebalanced test set would mask false-positive dynamics entirely.

D. Detector Architectures

Four edge detectors are benchmarked at their lightweight nano variant, matching micro-edge compute constraints: (1) **YOLO11n** [9], an anchor-free CNN with C3k2 convolutional backbone and C2PSA neck self-attention; (2) **YOLO26n** [10], utilizing structural reparameterization blocks (RepConv) for feedforward edge inference; (3) **YOLO12n** [11], an attention-centric detector employing Area Attention (A2C2f) for linear computational complexity; and (4) **YOLOv10n** [20], introducing dual-label assignments for NMS-free real-time inference. Table I summarizes architectural specifications and measured inference latency. All four models share lightweight capacity (2.3–2.6M parameters, 6.2–7.6 GFLOPs, and latency under 22 ms), eliminating parameter volume as a major confounding variable.

TABLE I
EVALUATED DETECTOR ARCHITECTURES (640×640 INPUT)

Detector	Params	FLOPs	Latency*	FPS*	Paradigm	Mechanism
YOLO11n	2.6 M	6.5 G	16.8 ms	59.6	Hybrid	C3k2 + C2PSA
YOLO26n	2.6 M	6.2 G	21.7 ms	46.1	Pure Conv	RepConv
YOLO12n	2.6 M	7.6 G	21.3 ms	47.0	Attn	A2C2f Area Attn
YOLOv10n	2.3 M	6.6 G	7.7 ms	129.2	NMS-Free	Dual-Label Head

*Latency and throughput measured on NVIDIA RTX 4060 Laptop GPU (PyTorch, batch=1, 640×640, averaged over 200 iterations after 50 warmups).

E. Experimental Configurations

1) Axis 1 (RQ1): Negative-Frame Ratio Sweep: Five training sets hold the positive core fixed at $|\mathcal{D}_{\mathrm{pos}}| = 2{,}401$ frames and sample strictly nested subsets of negative frames at uniform arithmetic 20%-point increments: 0% (2,401 total; 0 neg), 20% (3,001 total; 600 neg), 40% (4,001 total; 1,600 neg), 60% (6,003 total; 3,602 neg), and 80% (12,005 total; 9,604 neg). The 80% condition uses 9,604 of the 10,178 available training negatives; the remaining 574 are excluded to maintain exact 20-percentage-point arithmetic stepping. The negative subsets are strictly nested: $\mathcal{D}_{\mathrm{neg}}^{(0\%)} \subset \mathcal{D}_{\mathrm{neg}}^{(20\%)} \subset \cdots \subset \mathcal{D}_{\mathrm{neg}}^{(80\%)}$ using a deterministic random seed (seed=42). Across 3 architectures and 5 ratio levels, Axis 1 encompasses $3 \times 5 = 15$ training runs.

2) Axis 2 (RQ2): Hard-Negative Curation: For each architecture, a baseline detector f_{θ_0} trained on positive-only data runs inference across the 10,178-frame negative candidate pool. Frames yielding false positives above τ=0.25 are ranked by maximum confidence $s_{\max}(X)$, and the top N_{target} frames form the hard-mined set (backfilled deterministically from remaining negatives if needed). We evaluate hard-negative curation at the compute-efficient reference ratio $r_{\mathrm{ref}} = 40\%$ ($N_{\mathrm{target}} = 1{,}600$ negatives), which represents the compute-efficient compromise identified in Section V. Training set cardinality is strictly matched between random and curated conditions: $|\mathcal{D}_{\mathrm{train,hard}}^{(40\%)}| = |\mathcal{D}_{\mathrm{train,rand}}^{(40\%)}| = 2{,}401 + 1{,}600 = 4{,}001$ frames. Axis 2 adds $3 \times 2 = 6$ runs, establishing a total of **21 planned runs**.

F. Training Protocol

All models fine-tune from COCO-pretrained checkpoints at 640×640, seed=42. Training uses FP32 precision to circumvent a Windows cuBLAS FP16 driver fault on Ada Lovelace GPUs. Table II documents the frozen configuration (held identical across all models to isolate spatial aggregation). Training hardware: NVIDIA GeForce RTX 4060 Laptop GPU (8 GB VRAM). Target deployment hardware: NVIDIA Jetson Orin Nano (8 GB).

To guarantee reproducible execution: (1) **Batch Execution:** physical batch 16 without gradient accumulation;

TABLE II
UNIFIED FROZEN TRAINING HYPERPARAMETERS

Hyperparameter	Frozen Configuration
Input Resolution	640×640 (FP32 precision)
Physical / Effective Batch	16 / 16 (no accumulation)
Training Epochs	100 epochs
Optimizer	AdamW (lr_0=0.00125)
Weight Decay	0.0005
Augmentation Cooldown	Epoch 90 (`close_mosaic=10`)
Fixed Random Seed	42

Parity across models isolates feature aggregation design.

(2) **Precision:** FP32 (`amp=False`) avoids cuBLAS GEMM faults on Windows Ada Lovelace GPUs; (3) **Augmentation:** mosaic augmentation is disabled for the final 10 epochs (`close_mosaic=10`); (4) **Parity:** optimizer schedules and augmentations are held invariant across splits.

Training Compute Note: All ratio conditions train for a fixed 100 epochs regardless of dataset size. Because dataset sizes range from 2,401 frames (r=0%) to 12,005 frames (r=80%), higher-ratio models see approximately $5\times$ more total gradient updates. This design choice—standard in transfer-learning practice where epoch count, not step count, is the natural budget unit—means that the observed ratio effects co-vary with total training compute. We cannot separate the contribution of negative-frame content from additional optimization steps within this protocol; ratio-effect claims should therefore be interpreted as correlational. A step-matched ablation is left for future work.

IV. EXPERIMENTAL RESULTS

A. Negative-Frame Ratio Sensitivity (RQ1)

Table III reports detection accuracy and deployment cost across all ratio-sweep configurations on the held-out test set (1,572 frames, 80.9% negative prevalence).

The results across all four architectures reveal two consistent patterns. First, the *net direction* of false-positive suppression from 0% to 80% is universal: YOLO26n achieves a $31.5\times$ reduction (126→4 raw FPs), YOLOv10n $14.5\times$ (87→6 raw FPs, -93.1%), YOLO11n $12.7\times$ (76→6), and YOLO12n $7.3\times$ (66→9). At 80% negatives, all four models converge to a narrow false-positive band (4–9 raw detections per 1,272 negative test frames, 3.14–7.08 $\mathrm{FP}/1\mathrm{k}$). Across all architectures, the initial $0\% \rightarrow 20\%$ step provides the sharpest marginal reduction (YOLO26n: -84.1%; YOLOv10n: -65.5%; YOLO11n: -47.4%; YOLO12n: -31.8%), demonstrating that even modest negative-frame exposure stabilizes background activations.

Second, peak detection sensitivity (mAP_{50}) exhibits architecture-specific optima: YOLOv10n peaks at r=60% (0.9749), YOLO26n at r=40% (0.9830), while YOLO11n (0.9933) and YOLO12n (0.9925) maintain top sensitivity at r=80%. In all four detectors, high negative prevalence sustains strong localization ($\mathrm{mAP}_{50:95} \geq 0.738$), confirming that background supervision curbs false alarms without degrading learned spatial representations. Fig. 1a and Fig. 1b visualize these trajectories.

TABLE III
NEGATIVE-FRAME RATIO SWEEP ACROSS ARCHITECTURES (RQ1)

Det.	r	mAP_{50}	$\mathrm{mAP}_{50:95}$	P	R	$\mathrm{FP}/1\mathrm{k}$	Raw FP
YOLO11n	0%	0.9769	0.7538	0.8806	0.9741	59.75	76
	20%	0.9878	0.7518	0.9202	0.9728	31.45	40
	40%	0.9932	0.7699	0.9376	0.9839	24.37	31
	60%	0.9881	**0.7747**	0.9248	**0.9920**	12.58	16
	80%	0.9933	0.7606	**0.9644**	0.9766	**4.72**	6
YOLO26n	0%	0.9176	0.7095	0.8649	0.8988	99.06	126
	20%	0.9744	0.7632	0.9390	0.9785	15.72	20
	40%	**0.9830**	0.7555	0.9411	0.9653	5.50	7
	60%	0.9660	0.7594	**0.9587**	0.9086	10.22	13
	80%	0.9736	**0.7671**	0.9217	**0.9794**	**3.14**	4
YOLO12n	0%	0.9899	0.7653	0.9096	0.9840	51.89	66
	20%	0.9830	0.7481	0.9147	0.9837	35.38	45
	40%	0.9882	**0.7772**	0.9150	0.9959	18.08	23
	60%	0.9888	0.7766	0.9348	**0.9960**	9.43	12
	80%	**0.9925**	0.7529	**0.9621**	0.9767	**7.08**	9
YOLOv10n	0%	0.9196	0.7124	0.9016	0.8361	68.40	87
	20%	0.9459	0.7253	**0.9264**	0.9083	23.58	30
	40%	0.9615	0.7428	0.8918	**0.9765**	14.15	18
	60%	**0.9749**	**0.7525**	0.9221	0.9758	**4.72**	6
	80%	0.9524	0.7384	0.9010	0.9024	**4.72**	6

Single seed (seed=42); 1,272 negative test frames. $\mathrm{FP}/1\mathrm{k}$ at τ=0.25; P/R at max-F_1 threshold. Raw FP = absolute false-positive detections on the 1,272 negative frames. Bold marks best per architecture where the gap to the next-best exceeds 0.005 (accuracy) or represents the column minimum (FP/1k).

TABLE IV
HARD-NEGATIVE MINING VS. RANDOM SAMPLING AT MATCHED CARDINALITY (r_{ref}=40%, RQ2)

Det.	Curation	mAP_{50}	$\mathrm{mAP}_{50:95}$	$\mathrm{FP}/1\mathrm{k}$	$\%\Delta$	Mined	Fill
YOLO11n	Random	0.9932	0.7699	24.37	-71.0%	596	1004
	Hard-Mined	0.9946	0.7786	7.08			
YOLO26n	Random	0.9830	0.7555	5.50	0.0%	948	652
	Hard-Mined	0.9845	0.7585	5.50			
YOLO12n	Random	0.9882	0.7772	18.08	-65.2%	548	1052
	Hard-Mined	0.9933	0.7838	6.29			

N_{target}=1,600 negatives per architecture. **Mined**: frames flagged as FPs by the baseline (r=0%) model at τ=0.25. **Fill**: deterministically backfilled from remaining negatives to reach N_{target}. Mining purity = Mined / 1600.

B. Hard-Negative Curation vs. Random Sampling (RQ2)

Table IV benchmarks hard-negative mining against uniform random sampling at the compute-efficient reference ratio $r_{\mathrm{ref}} = 40\%$, with strictly matched cardinality (N=4,001 frames).

Table IV and Fig. 1c compare hard-negative mining against uniform random sampling. The attention-equipped models show large FP reductions: YOLO11n -71.0% (24.37 $\rightarrow$ 7.08 $\mathrm{FP}/1\mathrm{k}$) and YOLO12n -65.2% (18.08 $\rightarrow$ 6.29 $\mathrm{FP}/1\mathrm{k}$), with concurrent $\mathrm{mAP}_{50:95}$ improvements. YOLO26n shows an identical false-positive rate under both conditions (5.50 $\mathrm{FP}/1\mathrm{k}$, 7 raw detections on identical cabin outliers).

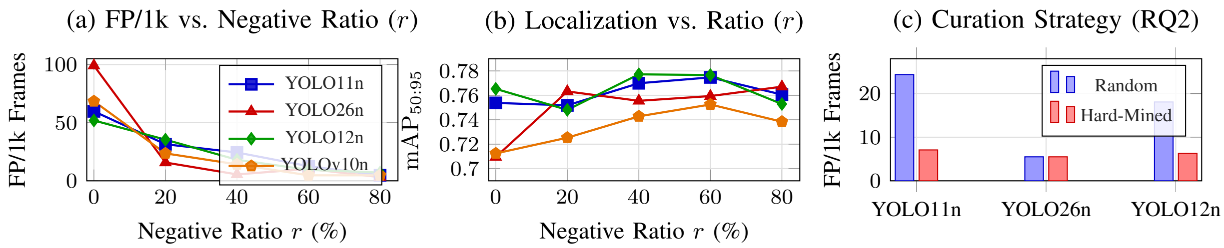

Fig. 1. Cross-architecture empirical findings: (a) False-positive rates decrease in aggregate (0%→80%) across all models, converging to 4–9 raw FPs at the 80% condition (YOLO26n shows a non-monotonic bump at r=60%); (b) Bounding-box localization ($\mathrm{mAP}_{50:95}$) remains robust across negative dilution (> 0.71), suggesting false-alarm suppression does not degrade detection accuracy; (c) Hard-negative curation at matched cardinality (N=4,001) substantially suppresses false alarms for attention-equipped architectures (YOLO11n: -71.0%, YOLO12n: -65.2%) while yielding no change for pure convolutions (YOLO26n: 0.0%)—though mining-pool purity differs across architectures (see Section IV-B).

Mining-pool purity and a complicating observation. The "Mined" and "Fill" columns in Table IV reveal that mining purity varies inversely with the observed benefit. YOLO26n's mined set is 59.2% genuinely curated (948/1,600)—the purest of the three—yet shows zero FP gain. YOLO11n (37.2% pure) and YOLO12n (34.2% pure) contain majority backfill yet show the largest reductions. If curation purity alone were the driver, the pattern should run the other way. We identify two candidate explanations, neither fully resolved within this single-seed protocol: (1) the backfilled fraction in YOLO11n and YOLO12n's mined sets partially overlaps with negative content already present in the random-40% baseline, meaning the measured effect conflates genuine curation with incidental re-sampling; (2) YOLO26n's convolution-only backbone may already saturate its background-suppression capacity at r=40% random (it reaches only 7 raw FPs at that point), leaving no headroom for mining to improve upon. The title's framing—"Curate for Attention, Sample for Convolutions"—captures the observed differential response but should be understood as a preliminary empirical pattern, not a confirmed causal mechanism. Resolving whether the driver is architectural receptive-field structure, mining purity, or baseline headroom requires a controlled experiment equalizing mined fractions across architectures, ideally with multiple seeds.

C. Operational Deployment Cost Translation

Table V translates empirical $\mathrm{FP}/1\mathrm{k}$ values into estimated hourly nuisance alerts ($\mathcal{A}_h$) via (4) across three standard camera streaming frequencies.

This comparison highlights a foundational distinction in edge AIoT system engineering: while raw inference latency is architecture-bound (all three models executing within 16.8–21.7 ms on edge hardware, Table I), operational false-alarm rate is curation-bound (varying by up to 31.5× depending on training-data composition). An ultra-fast detector running at ∼46–60 FPS is practically unusable if it triggers thousands of spurious alerts per hour. Under 0% baseline training, all three detectors produce between 756 and 1,443 nuisance alerts per hour at a modest 5 FPS, and between 4,534 and 8,655 alerts per hour at 30 FPS. Training with 80% negative frames reduces these projections to 46 alerts/hour for YOLO26n, 69 alerts/hour for YOLO11n, and 103 alerts/hour for YOLO12n at 5 FPS. Each suppressed detection preserves edge compute, prevents unnecessary cellular uploads, avoids SoC thermal throttling, and preserves operator trust.

TABLE V
PROJECTED NUISANCE ALERTS PER HOUR ($\mathcal{A}_h$) ACROSS FRAME RATES

Detector	Training Split	5 FPS	15 FPS	30 FPS
YOLO11n	0% Neg. Baseline	870	2,610	5,220
	Balanced Split (40%)	355	1,065	2,129
	80% Neg. Ratio	69	206	412
YOLO26n	0% Neg. Baseline	1,443	4,328	8,655
	Balanced Split (40%)	80	240	481
	80% Neg. Ratio	**46**	**137**	**274**
YOLO12n	0% Neg. Baseline	756	2,267	4,534
	Balanced Split (40%)	263	790	1,580
	80% Neg. Ratio	103	309	619

Modeled via (4) using operational negative prevalence p_{neg}=0.809. Minimum alert rates in bold.

V. DISCUSSION

A. Practical Guidelines for AIoT Data Curation

Our empirical findings offer two operational recommendations:

1) False-Alarm Minimization (Safety-Critical AIoT): For systems where false alarms degrade user trust or incur uplink costs, training at the 80% negative ratio is globally superior across all three architectures. YOLO26n provides the lowest absolute false-alarm rate (3.14 $\mathrm{FP}/1\mathrm{k}$, yielding 46 alerts/hr at 5 FPS), making pure reparameterized convolutions well-suited for high-precision, low-power sensing.

2) Compute-Constrained Training and High Localization: When training budgets limit dataset size, a 40% negative ratio offers the best compromise. It retains $\geq 98.3\%$ of peak mAP_{50}, achieves top-tier localization ($\mathrm{mAP}_{50:95} = 0.7772$ for YOLO12n), and captures the bulk of false-alarm suppression with less than one-third the negative volume of the 80% split.

Second, while detection sensitivity (mAP_{50}) exhibits broad stability across negative ratios ($r \in [40\%, 80\%]$), false-positive suppression magnitudes diverge sharply by architecture: YOLO26n suppresses false alarms by 31.5×, YOLO11n by 12.7×, and YOLO12n by 7.3×. This divergence suggests that edge architects cannot decouple data curation from detector selection: models relying on attention mechanisms require higher negative densities to achieve comparable background suppression as pure convolutional backbones.

B. Limitations

Several methodological constraints contextualize our findings:

- **Training Step Confound:** Fixed 100-epoch training across variable dataset sizes (2,401 to 12,005 frames) means the 80% condition benefits from $\approx 5\times$ more gradient steps than the 0% baseline. Disentangling negative content from total optimization budget requires step-matched ablation studies.
- **Mining Purity Variations:** The hard-negative pools contained differing fractions of genuinely mined false positives (34.2%–59.2%) due to deterministic backfill to $N{=}1{,}600$, partially confounding curation response with filler composition.
- **Frame-Level Partitioning:** Consecutive frames within a recording session share cabin environment and lighting. Near-ceiling detection performance ($\mathrm{mAP}_{50} > 0.97$) across all splits may partly reflect temporal proximity correlation rather than cross-vehicle generalization; subject-disjoint multi-cohort evaluation is needed.
- **Foreground Class Imbalance:** The positive core is heavily dominated by `phone_use` (81.2%), meaning aggregate mAP primarily mirrors majority-class accuracy rather than rarer, safety-critical fatigue indicators (`yawning`, 5.3%).
- **Single-Seed Point Estimates:** Results stem from a single deterministic seed (seed=42), meaning small count differences (e.g., 4 vs. 7 raw false alarms) cannot be separated from stochastic training noise without multi-seed confidence intervals.

VI. Conclusion

This paper investigated the interaction between negative-frame training curation and edge detector architectures across matched-capacity ($\approx 2.6\,\mathrm{M}$ parameter) models in continuous AIoT sensing. Our empirical findings demonstrate that while models exhibit diverse suppression trajectories across negative ratios—with pure reparameterized convolution showing steep initial suppression and attention-equipped variants showing sustained sensitivity—all three architectures converge to a narrow false-positive band (4–9 raw false detections on 1,272 test frames) under high negative prevalence ($r{=}80\%$). Because on-device latency remains strictly architecture-bound (16.8–21.7 ms), operational nuisance alert rates are primarily governed by training-data composition rather than architectural choice. For edge AIoT practitioners, the core takeaway is practical: curate negative frames to reflect real-world operational background dominance first, and select detector backbones based on hardware latency targets rather than relying on architectural inductive biases alone to suppress false alarms.

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